

CS 4530: Fundamentals of Software Engineering

Module 06: React Hook Patterns

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Learning Objectives for this Module

- By the end of this module, you should be able to:
 - Explain the basic use cases for useEffect
 - Explain when a useEffect is executed, and when its return value is executed
 - Construct simple custom hooks and explain why they are useful.
 - Be able to explain the three core steps of a test (assemble, act, assess) can map to UI component testing
 - Become familiar with Best practices of UI/UX design

Lesson 6.1 useEffect

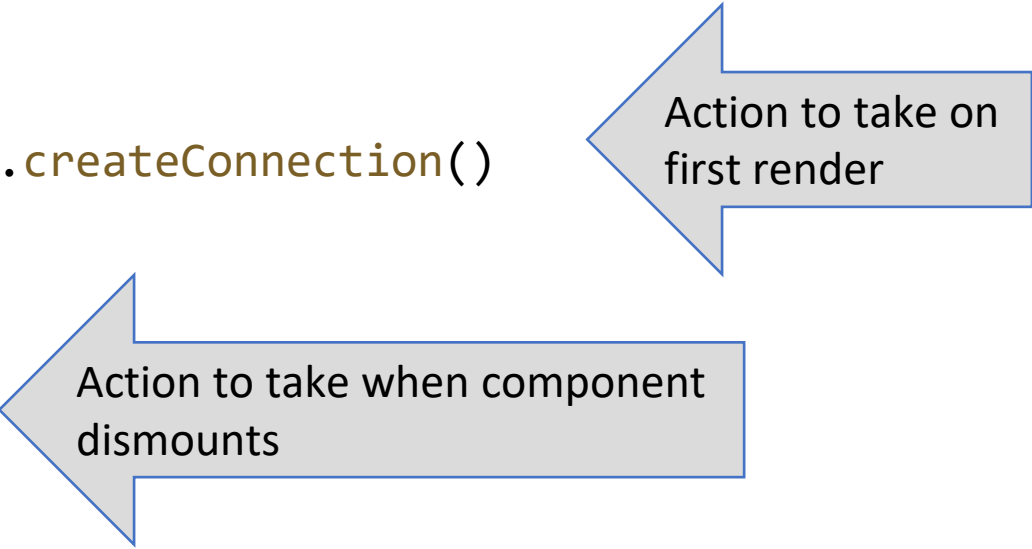
useEffect is a mechanism for synchronizing a component with an external system

```
import { clockServer } from './clock.js';

function ClockClient() {

  useEffect(() => {
    const connection = clockServer.createConnection()
    connection.connect();

    return () => {
      connection.disconnect();
    };
  }, []);
  // ...
}
```



Action to take on first render

Action to take when component dismounts



Empty array says: do this on first render only

<https://react.dev/reference/react/useEffect>

An external system means any piece of code that's not inside your React component

- An event in the lifecycle of a component, like `render`.
- A timer managed with `setInterval` and `clearInterval`
- An event subscription like a chat server (sockets?)
- A call to fetch data from an external API or web site
- An external animation library
- A piece of business logic in an app that is external to your component

A real example: a display that connects to a self-ticking clock

src/Components/SimpleClockDisplay.tsx

```
export default function ClockDisplay(props: {
  name: string, key: number,
  clock: IClock,
  handleDelete: () => void,
  handleAdd: () => void,
}) {
  {
    const [localTime, setLocalTime] = useState(0)
    const incrementLocalTime = () => setLocalTime(localTime => localTime + 1)
    const clock = props.clock

    useEffect(() => {
      const listener1 = () => { incrementLocalTime() }
      clock.addListener(listener1)
      return () => {
        clock.removeListener(listener1)
      }
    }, [])
```

The parent provides the clock

On first render, add this listener to the clock

On dismount, remove the listener.

Display logic will come later...

Our app will have three displays of the clock

```
import { Heading, VStack } from '@chakra-ui/react';
import ClockDisplay from './ClockDisplay.tsx';
import SingletonClockFactory from '../../Classes/SingletonClockFactory.ts';

export default function App() {
  const clock = SingletonClockFactory.getInstance(1000);
  clock.start()

  return (
    <VStack>
      <Heading>Three Clocks</Heading>
      <ClockDisplay key={1} name={'Clock A'} clock={clock} />
      <ClockDisplay key={2} name={'Clock B'} clock={clock} />
      <ClockDisplay key={3} name={'Clock C'} clock={clock} />
    </VStack>
  );
}
```

The clock itself is simple

```
// the listener does not expect an argument
type Listener = () => void;
```

```
class Clock implements IClock {
  // the actual clock state.
  // it runs all the time, but only notifies
  // listeners when this._running is true.
```

```
  private _createTimer() {
    setInterval(() => {
      if (this._running) {
        this._notifyAll();
      }
    }, this._interval);
  }
```

```
  private _running: boolean = false;
```

```
// the arrow-function idiom guarantees
// that 'this' is bound statically.
```

```
  public start = () => {
    this._running = true;
  };
```

```
  public stop = () => {
    this._running = false;
  };
```

```
  public constructor(interval: number) {
    this._interval = interval;
    this._createTimer();
    this.start();
  }
```


And we'll make the clock a singleton in the usual way

src/Classes/SingletonClockFactory.ts

```
export default class SingletonClockFactory {  
  
    private static theClock: Clock | undefined = undefined  
  
    private constructor () {SingletonClockFactory.theClock = undefined}  
  
    public static instance (interval:number) : Clock {  
        if (SingletonClockFactory.theClock === undefined) {  
            SingletonClockFactory.theClock = new Clock(interval)  
        }  
        return SingletonClockFactory.theClock  
    }  
}
```

Now let's look at <ClockDisplay>

```
export default function ClockDisplay(props: ClockDisplayProps) {
  const [localTime, setLocalTime] = useState(0);
  const incrementLocalTime = () => {
    setLocalTime(localTime => localTime + 1);
  };

  useEffect(() => {
    const listener1 = () => {
      incrementLocalTime();
    };
    props.clock.addListener(listener1);
    console.log(`ClockDisplay ${props.name} is mounting`);
    return () => {
      console.log('ClockDisplay ' + props.name + ' is unmounting');
      props.clock.removeListener(listener1);
    };
  }, [props.clock, props.name]);
}
```

Clock Display: display logic

```
return (  
  <HStack>  
    <Box>Clock: {props.name}</Box>  
    <Box>Time = {localTime}</Box>  
    <Box>nlisteners = {clock.nListeners}</Box>  
    <Button aria-label={'start'} onClick={() => clock.start()}>  
      Start  
    </Button>  
    <Button aria-label={'stop'}  onClick={() => clock.stop()}>  
      Stop  
    </Button>  
  </HStack>  
)  
);  
}
```

Clock: Clock A Time = 11 nlisteners = 3



Clock: Clock B Time = 11 nlisteners = 3



Clock: Clock C Time = 11 nlisteners = 3



Elements Console Sources >> ⚙️ ⋮ ✕

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No Issues

ClockDisplay Clock A is mounting [SimpleClockDisplay.tsx:24](#)

ClockDisplay Clock B is mounting [SimpleClockDisplay.tsx:24](#)

ClockDisplay Clock C is mounting [SimpleClockDisplay.tsx:24](#)



useEffect's Dependencies Control Its Execution

- `useEffect` takes an optional array of dependencies
- The effect is only executed if one or more of the values in the dependency change (e.g. by a setter)
- Special Cases:
 - `[]` means run only on first render
 - No argument means run on every render

Example (Part 1)

```
export default function App() {
  const [i, setI] = useState(0);
  const [j, setJ] = useState(0);

  // runs only on first render.
  useEffect(() => {
    console.log('useEffect #1 is run only on first render');
  }, []);

  useEffect(() => {
    console.log('useEffect #2I is run when i changes');
  }, [i]);

  useEffect(() => {
    console.log('useEffect #2J is run when j changes');
  }, [j]);

  useEffect(() => {
    console.log('useEffect #2IJ is run when either i or j
changes');
  }, [i, j]);

  // runs on every render
  useEffect(() => {
    console.log('useEffect #3 is called on every render');
  });
}
```

```
function onClickI() {
  console.log('Clicked i!');
  setI(lastI => lastI+1);
  // bound variable name doesn't matter, of course
  // setI(i+1) is a bug, because value of i may not
  // be current.
}

function onClickJ() {
  console.log('Clicked j!');
  setJ(j => j+1);
}

return (
  <VStack>
    <Heading>useEffect demo #1IJ</Heading>
    <Text> i is {i} </Text>
    <Button onClick={onClickI}>Increment i</Button>
    <Text> j is {j} </Text>
    <Button onClick={onClickJ}>Increment j</Button>
  </VStack>
);
```

Demo

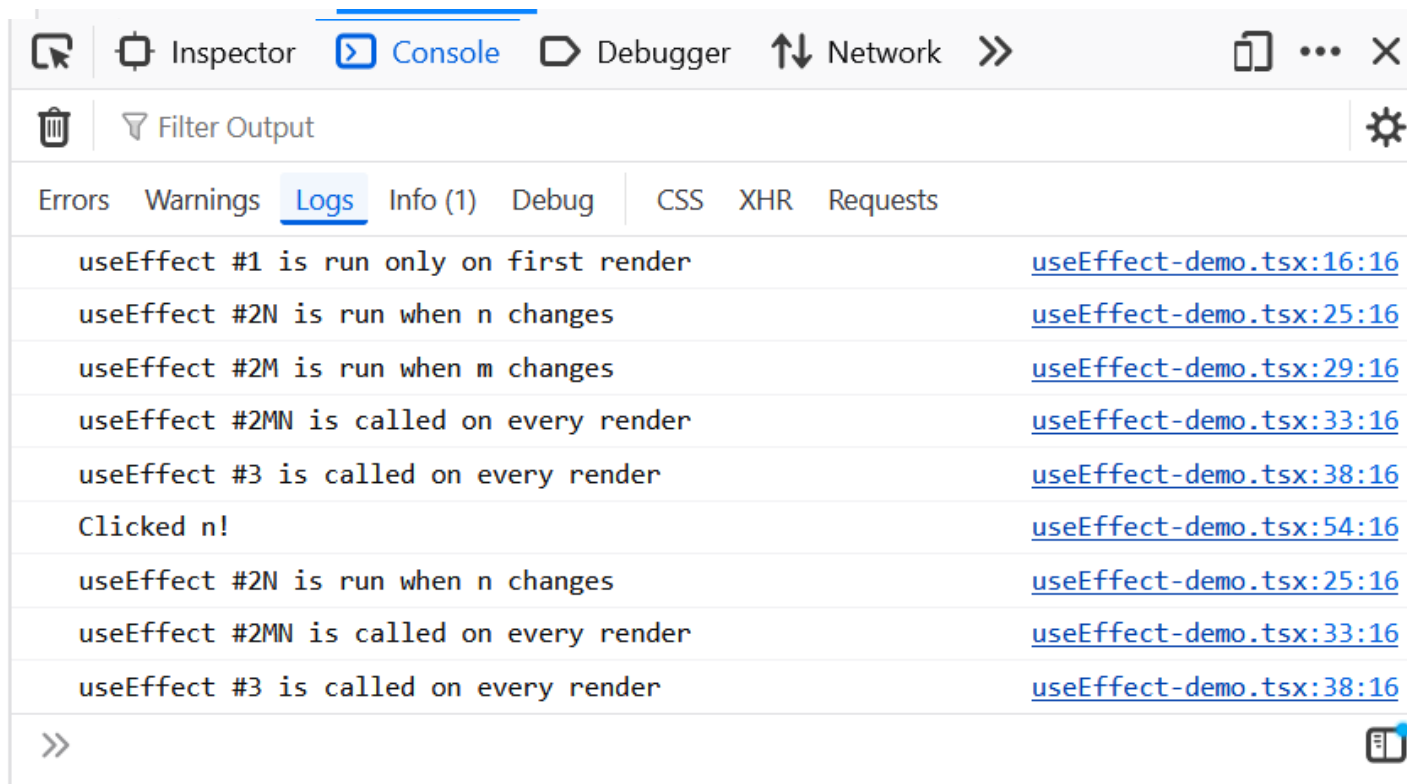
useEffect demo #1

n is 1

Increment n

m is 0

Increment m



The screenshot shows the Chrome DevTools console with the 'Console' tab selected. The logs are as follows:

Log Message	Source
useEffect #1 is run only on first render	useEffect-demo.tsx:16:16
useEffect #2N is run when n changes	useEffect-demo.tsx:25:16
useEffect #2M is run when m changes	useEffect-demo.tsx:29:16
useEffect #2MN is called on every render	useEffect-demo.tsx:33:16
useEffect #3 is called on every render	useEffect-demo.tsx:38:16
Clicked n!	useEffect-demo.tsx:54:16
useEffect #2N is run when n changes	useEffect-demo.tsx:25:16
useEffect #2MN is called on every render	useEffect-demo.tsx:33:16
useEffect #3 is called on every render	useEffect-demo.tsx:38:16

When is the cleanup function executed?

- In general, the cleanup function is executed sometime before the next time the hook is run.
- For the first-time-only case, this means when the component is dismantled.
- Let's look at useEffect demo again, this time with noisy cleanups.

Demo

```
function cleanup(message: string) {  
  return () => {  
    console.log('cleanup: ' + message);  
  };  
}  
  
export default function App() {  
  const [i, setI] = useState(0);  
  const [j, setJ] = useState(0);  
  
  // runs only on first render.  
  useEffect(() => {  
    console.log('useEffect #1 is run only on first render');  
    return cleanup('useEffect #1');  
  }, []);  
  
  useEffect(() => {  
    console.log('useEffect #2I is run when i changes');  
    return cleanup('useEffect #2I');  
  }, [i]);  
  
  // etc.
```

useEffect demo with CleanUps

n is 1

Increment n

m is 0

Increment m

The screenshot shows a web browser's developer console with the 'Console' tab selected. The console displays a series of log messages related to the useEffect hook and its cleanup functions. The messages are as follows:

- useEffect #1 is run only on first render [...Effect-demoWithCleanUps.tsx:20:16](#)
- useEffect #2N is run only when n changes [...Effect-demoWithCleanUps.tsx:25:16](#)
- useEffect #2M is run when m changes [...Effect-demoWithCleanUps.tsx:30:16](#)
- useEffect #2MN is called when m or n changes [...Effect-demoWithCleanUps.tsx:36:16](#)
- useEffect #3 is called on every render [...Effect-demoWithCleanUps.tsx:42:16](#)
- Clicked n! [...Effect-demoWithCleanUps.tsx:54:16](#)
- cleanup: useEffect #2N [...Effect-demoWithCleanUps.tsx:10:57](#)
- cleanup: useEffect #2MN [...Effect-demoWithCleanUps.tsx:10:57](#)
- cleanup: useEffect #3 [...Effect-demoWithCleanUps.tsx:10:57](#)
- useEffect #2N is run only when n changes [...Effect-demoWithCleanUps.tsx:25:16](#)
- useEffect #2MN is called when m or n changes [...Effect-demoWithCleanUps.tsx:36:16](#)
- useEffect #3 is called on every render [...Effect-demoWithCleanUps.tsx:42:16](#)

The console interface includes a toolbar at the top with icons for Inspector, Console, Debugger, Network, and a search icon. Below the toolbar is a 'Filter Output' section. The log messages are categorized into Errors, Warnings, Logs, Info (1), Debug, CSS, XHR, and Requests. The 'Logs' tab is currently selected, showing the log messages. The console also has a 'Clear' button (trash icon) and a 'Settings' button (gear icon).

Lesson 6.2 Custom Hooks

Custom Hooks

- REACT lets us combine `useState` and `useEffect` to build custom hooks.
- Custom Hooks let us separate business logic from display logic

Example: useClock

```
import { useEffect } from 'react';
import SingletonClockFactory from '../Classes/SingletonClockFactory.ts';
import { type IClock } from '../Interfaces/IClock.ts';

// takes a listener function as an argument
// returns the clock object
export function useClock(listener1: () => void): IClock {
  const clock = SingletonClockFactory.getInstance(1000);
  useEffect(() => {
    clock.addListener(listener1);
    return () => {
      clock.removeListener(listener1);
    };
  }, [clock, listener1]);
  return clock;
}
```

Using useClock

```
import { useClock } from '../Hooks/useClock';

export function ClockDisplay(props: {
  name: string, key: number,
  handleDelete: () => void, handleAdd: () => void,
  noisyDelete?: boolean

}) {
  const [localTime, setLocalTime] = useState(0)
  const incrementLocalTime = () => setLocalTime(localTime => localTime + 1)
  const clock:IClock = useClock(incrementLocalTime)

  return (
    <HStack>
      <Box>Clock: {props.name}</Box>
      <Box>Time = {localTime}</Box>
      <Box>nlisteners = {clock.nListeners}</Box>
      <IconButton aria-label={'delete'} onClick={props.handleDelete} icon={<AiOutlineDelete />} />
      <IconButton aria-label={'add'} onClick={props.handleAdd} icon={<AiOutlinePlus />} />
    </HStack>
  )
}
```

A somewhat larger example: ToDoList

```
export default function ToDoApp () {  
  const [todoList, setTodolist] = useState<ToDoItem[]>([])  
  const [itemKey, setItemKey] = useState<number>(0) // first unused key  
  
  function handleAdd (title:string, priority:string) {  
    if (title === '') {return} // ignore blank button presses  
    setTodolist(todoList.concat({title: title, priority: priority, key: itemKey}))  
    setItemKey(itemKey + 1)  
  }  
  
  function handleDelete(targetKey:number) {  
    const newList = todoList.filter(item => item.key !== targetKey)  
    setTodolist(newList)  
  }  
  
  return (  
    <VStack>  
      <Heading>TODO List</Heading>  
      <ToDoItemEntryForm onAdd={handleAdd}/>  
      <ToDoListDisplay items={todoList} onDelete={handleDelete}/>  
    </VStack>  
  )  
}
```



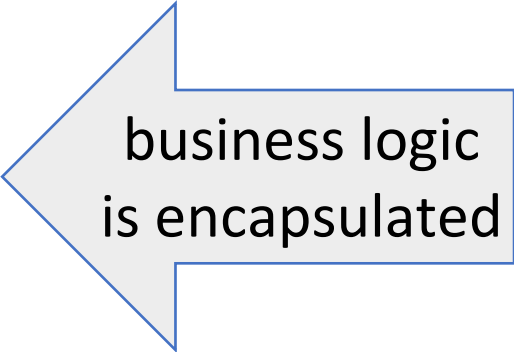
business logic



display logic

Refactoring ToDoList

```
export default function ToDoApp () {  
  
  const {todoList, handleAdd, handleDelete} = useToDoItemList()  
  
  return (  
    <VStack>  
      <Heading>TODO List</Heading>  
      <ToDoItemEntryForm onAdd={handleAdd}/>  
      <ToDoListDisplay items={todoList} onDelete={handleDelete}/>  
    </VStack>  
  )  
}
```



business logic
is encapsulated

The hook encapsulates the business logic

```
export default function useToDoItemList () {  
  const [todoList,setTodolist] = useState<ToDoItem[]>([])  
  const [itemKey,setItemKey] = useState<number>(0)    // first unused key  
  
  function handleAdd (title:string, priority:string) {  
    if (title === '') {return}    // ignore blank button presses  
    setTodolist(todoList.concat({title: title, priority: priority, key: itemKey}))  
    setItemKey(itemKey + 1)  
  }  
  
  function handleDelete(targetKey:number) {  
    const newList = todoList.filter(item => item.key !== targetKey)  
    setTodolist(newList)  
  }  
  
  return {todoList: todoList, handleAdd: handleAdd, handleDelete: handleDelete}  
}
```

The hook is like a class managing a piece of state

```
export default function useToDoItemList () {  
  const [todoList, setTodolist] = useState<ToDoItem[]>([])  
  const [itemKey, setItemKey] = useState<number>(0) // first unused key  
  
  function handleAdd (title:string, priority:string) {  
    if (title === '') {return} // ignore blank button presses  
    setTodolist(todoList.concat({title: title, priority: priority, key: itemKey}))  
    setItemKey(itemKey + 1)  
  }  
  
  function handleDelete(targetKey:number) {  
    const newList = todoList.filter(item => item.key !== targetKey)  
    setTodolist(newList)  
  }  
  
  return {todoList: todoList, handleAdd: handleAdd, handleDelete: handleDelete}  
}
```

handleAdd and handleDelete
are the only methods for
manipulating the state

The hook's state becomes part of its user's state.

```
export default function useToDoItemList () {  
  const [todoList, setTodolist] = useState<ToDoItem[]>([])  
  const [itemKey, setItemKey] = useState<number>(0) // first unused key  
  
  function handleAdd (title:string, priority:string) {  
    if (title === '') {return} // ignore blank button presses  
    setTodolist(todoList.concat({title: title, priority: priority, key: itemKey}))  
    setItemKey(itemKey + 1)  
  }  
  
  function handleDelete(targetKey:number) {  
    const newList = todoList.filter(item => item.key !== targetKey)  
    setTodolist(newList)  
  }  
  
  return {todoList: todoList, handleAdd: handleAdd, handleDelete: handleDelete}  
}
```

calling these setters redisplay
the whole component

The Rules of Hooks

1. Only call hooks at the top level

- Not within loops, inside conditions, or nested functions
- Rationale: The order of hooks called must always be the same each time a component renders

2. Only call hooks from React Components or Custom Hooks

- Not from any other helper methods or classes
- Rationale: React must know the component that the call to the hook is associated with

```
export function LikeButton() {  
  const [isLiked, setIsLiked] = useState(false);  
  const [count, setCount] = useState(0);  
  ...  
}
```

React knows which `useState` is which by tracking calls to them from components in the render tree

We Use Two ESLint Rules for React Hooks

- You should not violate the rules of hooks. These linter plugins help detect violations
- React-hooks/rules-of-hooks
 - Enforces that hooks are only called from React functional components or custom hooks
- React-hooks/exhaustive-deps
 - Enforces that all variables used in useEffects are included as dependencies

Putting it All Together

- In the previous examples, we learned
 - how to introduce side effects
 - how to create our own hooks
- We can use these concepts to make React components interact with a remote system (e.g., REST services).

Example (1): Interacting With a REST-based Server

```
export default function RandomNASA() {  
  const [image, setImage] = useState<null | string>(null);  
  const [error, setError] = useState<unknown>(null);
```

```
  useEffect(() => {  
    fetch("https://api.nasa.gov/planetary/apod?api_key=DEMO_KEY")  
      .then((response) => response.json())  
      .then((json) => {  
        const data = z.object({ url: z.string() }).parse(json);  
        setImage(data.url);  
      })  
      .catch((err) => setError(err));  
  }, []);
```

This function runs *after* the fetch returns

And this function runs after *that*

This function runs if there's an error in the previous

```
  return error !== null ? (  
    `Could not retrieve image: ${error}`  
  ) : image !== null ? (  
    <img src={image} />  
  ) : (  
    `Loading...`  
  );  
}
```

Example (2, 3): Setting Up and Cleaning Up a Chat Subscription or Showing Unread Messages

- `useEffect(() => {
 const socket = new WebSocket('wss://api.randomsite.com/chat');`

```
  socket.onmessage = (event) => {  
    const message = JSON.parse(event.data);  
    setMessages((prev) => [...prev, message]);  
  };  
  return () => {  
    socket.close();  
  };  
}, []);
```

This allows you to subscribe to live chat messages

This function runs at cleanup to avoid memory leaks

- `useEffect(() => {
 document.title = `You have ${unreadMsgs} unread messages`;
}, [unreadMsgs]);`

This updates the browser tab title to show the number of unread messages or notifications

There are several other React Hooks that might be useful.

- React Context allows you to share state between deeply nested components more easily

```
import { useContext } from 'react';

function myButtonComponent() {
  const theme = useContext(ThemeContext);
  // ...
}
```

- useRef Hook lets you reference a value that's not needed for rendering
- useEffect Hook lets you extract non-reactive logic from your Effects into a reusable function.

Lesson 6.3 Testing your React components

Testing React components

- The AAA pattern ("Assemble/Act/Assess") still applies
- Need a test double for the React system
 - render components into a "virtual dom" or into a captive web browser
- We'll be using a package called "Playwright" to do this.

Recall: Most tests are in AAA form: Assemble/Act/Assess

```
test('addStudent should add a student to the database', () => {  
  // const db = new DataBase ()  
  expect(db.nameToIDs('blair')).toEqual([])  
  
  const id1 = db.addStudent('blair');  
  
  expect(db.nameToIDs('blair')).toEqual([id1])  
});
```

The diagram illustrates the AAA (Assemble/Act/Assess) test structure. It shows a code snippet with three main parts, each highlighted by a green callout box with a red arrow pointing to the corresponding code:

- Assemble (and check that you've assembled it)**: Points to the first `expect` statement: `expect(db.nameToIDs('blair')).toEqual([])`.
- Act (do the action that you are trying to test)**: Points to the `const id1 = db.addStudent('blair');` statement.
- Assess: check to see that the response is correct**: Points to the second `expect` statement: `expect(db.nameToIDs('blair')).toEqual([id1])`.

Playwright commands work on a “page object”

<code>.goto()</code>	Navigate to a URL. Many tests begin with this command.
<code>.getByLabel()</code>	Find form elements by their associated label text.
<code>.getByRole()</code>	Find elements by their ARIA role(button, textbox, etc) .
<code>.getByText()</code>	Find elements by their text content.
<code>.waitForURL()</code>	Wait for navigation to complete to a specific URL.
<code>.fill()</code>	Type text into an input field.

These will fail if the specified element does not exist

A typical playwright test

```
test('should allow an existing user to log in', async ({ page }) => {  
  await page.goto('/login');  
  
  await page.getByLabel('Username')  
    .fill("Frau Drei");  
  
  await page.getByLabel('Password',  
    { exact: true }).fill(password);  
  
  await page.getByRole('button',  
    { name: 'Log In' }).click();  
  
  await page.waitForURL('/');  
  await expect(page.getByText  
    ('signed in as Frau Drei')).toBeVisible();  
});
```

Assemble (navigate to the page)

Act (fill form and click login)

Assess: check to see that the response is correct

run with: npx playwright test

Lesson 6.4 Best Practices for UI Design

UI/UX Design principles

- The most commonly repeated UI/UX design principles across companies like [Apple](#), [Google Material Design](#), [Microsoft Design](#), [IBM Carbon Design System](#), and the Nielsen Norman Group are surprisingly consistent:
 - Consistency
 - Clarity
 - Accessibility
 - Visual hierarchy
 - Simplicity
 - User control
 - Efficiency
 - Predictability
 - Responsive/adaptive design
 - Feedback/system status

Learning Objectives for this Lesson

- By the end of this lesson, you should be able to:
 - Explain the basic use cases for useEffect
 - Explain when a useEffect is executed, and when its return value is executed
 - Construct simple custom hooks and explain why they are useful.
 - Be able to explain the three core steps of a test (assemble, act, assess) can map to UI component testing
 - Become familiar with Best practices of UI/UX design